

PROJECT REPORT

AI-Based Learning Platform with Smart Study Planner

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Personalized learning and intelligent planning designed to help students study smarter, stay organized, and achieve better outcomes.

Project Overview

Technology Stack

- React.js
- FastAPI
- MySQL
- JWT
- Postman
- Git & GitHub

Key Features

- Course Management
- Task Tracking
- Daily Study Planning
- Progress Analytics
- AI-Powered Recommendations

Problem Statement & Solution

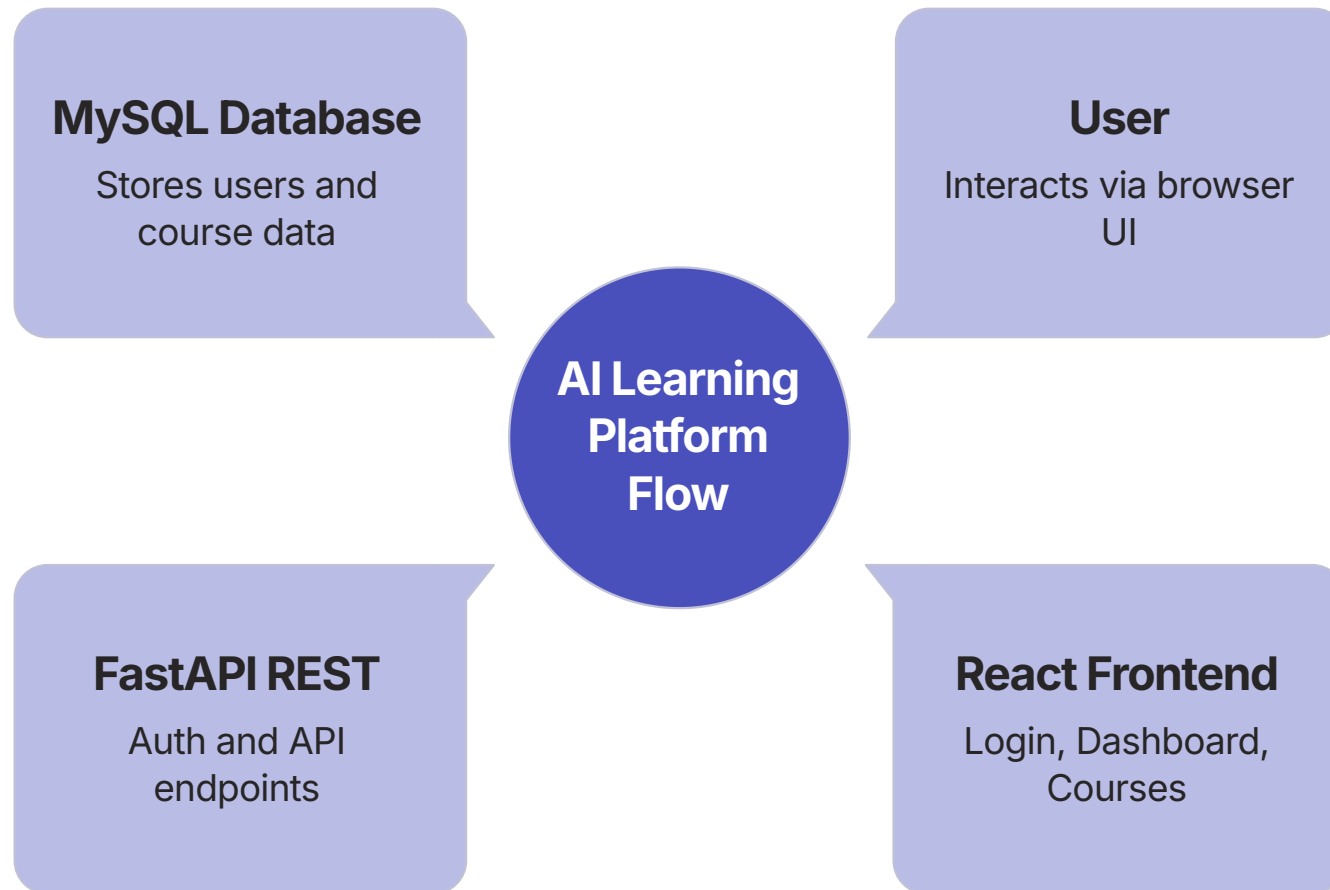
Problems

- Poor time management
- Missed assignment deadlines
- Lack of study planning
- Difficulty tracking progress
- No personalized learning guidance

Solution

- Intelligent course management
- Automated task tracking
- AI-powered study recommendations
- Real-time progress analytics
- Personalized learning guidance

System Architecture



This diagram illustrates the core components and data flow within the AI-Based Learning Platform.

Frontend (React.js)

- Login & Register pages
- Dashboard
- Courses, Tasks, Planner views

Backend (FastAPI)

- REST API endpoints
- Authentication logic
- Business logic

Database (MySQL)

- Users, Courses, Tasks
- Study Plans
- Analytics data

Core Modules

Authentication Module

- User registration
- User login
- JWT token generation

Course Module

- Create course
- View courses
- Delete courses

Task Module

- Create tasks
- Update task status
- Manage deadlines

Planner Module

- Daily study schedule
- Study slot allocation

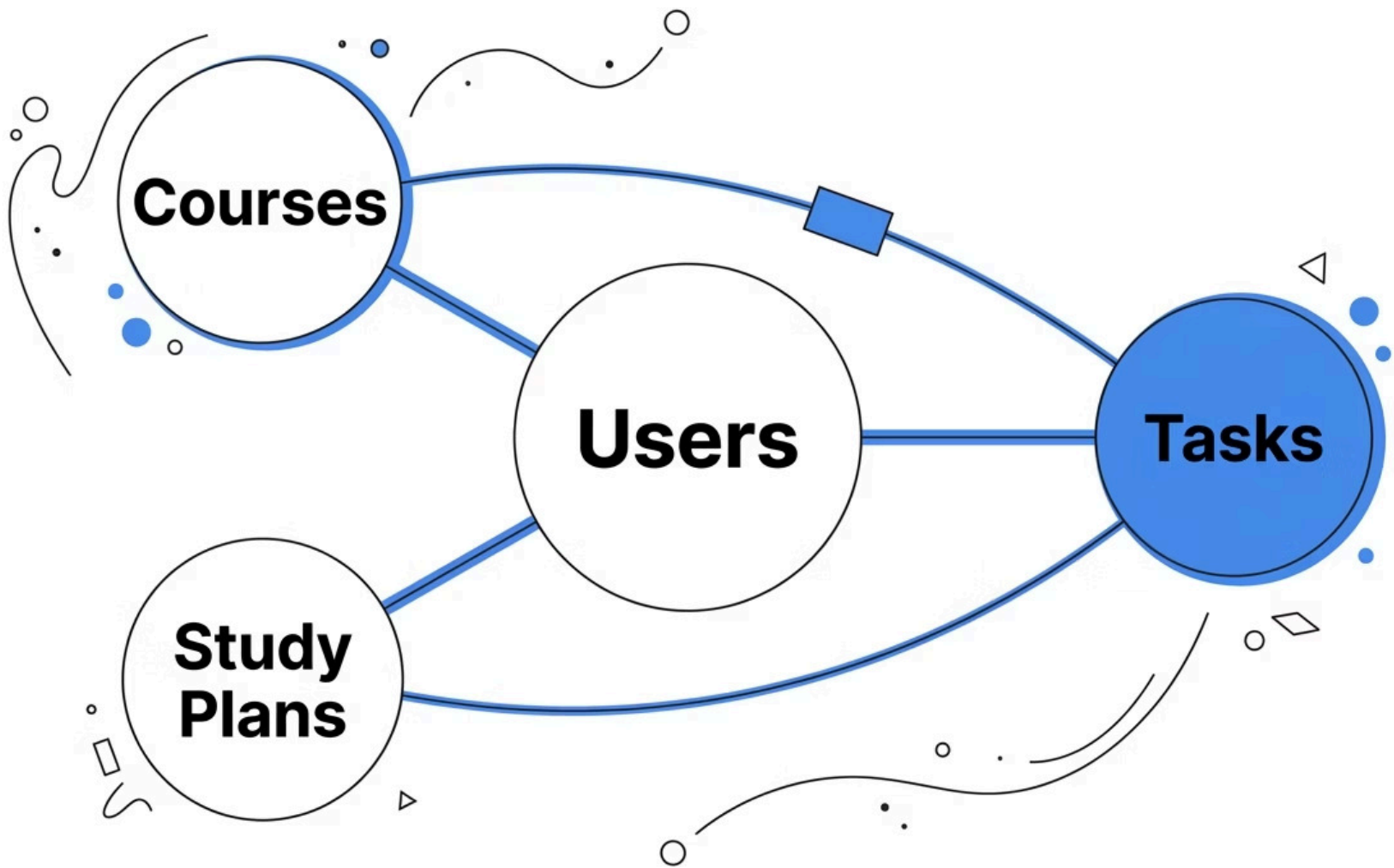
Analytics Module

- Progress calculation
- Completion statistics

AI Recommendation Module

- Task prioritization
- Study recommendations
- Performance suggestions

Database Design



The database is designed with a clear relational structure to manage user data, courses, tasks, and study plans effectively. Below is a detailed breakdown of each table and its key relationships.

Users Table

- **id**: Unique identifier for the user.
- **name**: User's full name.
- **email**: User's email address (unique).
- **password**: Hashed password for security.
- **created_at**: Timestamp of user creation.

Courses Table

- **id**: Unique identifier for the course.
- **user_id**: Foreign key linking to the Users table.
- **course_name**: Name of the course.
- **description**: Detailed description of the course.
- **start_date**: Date when the course starts.
- **end_date**: Date when the course ends.

Tasks Table

- **id**: Unique identifier for the task.
- **user_id**: Foreign key linking to the Users table.
- **course_id**: Foreign key linking to the Courses table.
- **title**: Title of the task.
- **description**: Description of the task.
- **priority**: Priority level of the task (e.g., High, Medium, Low).
- **deadline**: Deadline for the task.
- **status**: Current status of the task (e.g., To Do, In Progress, Completed).

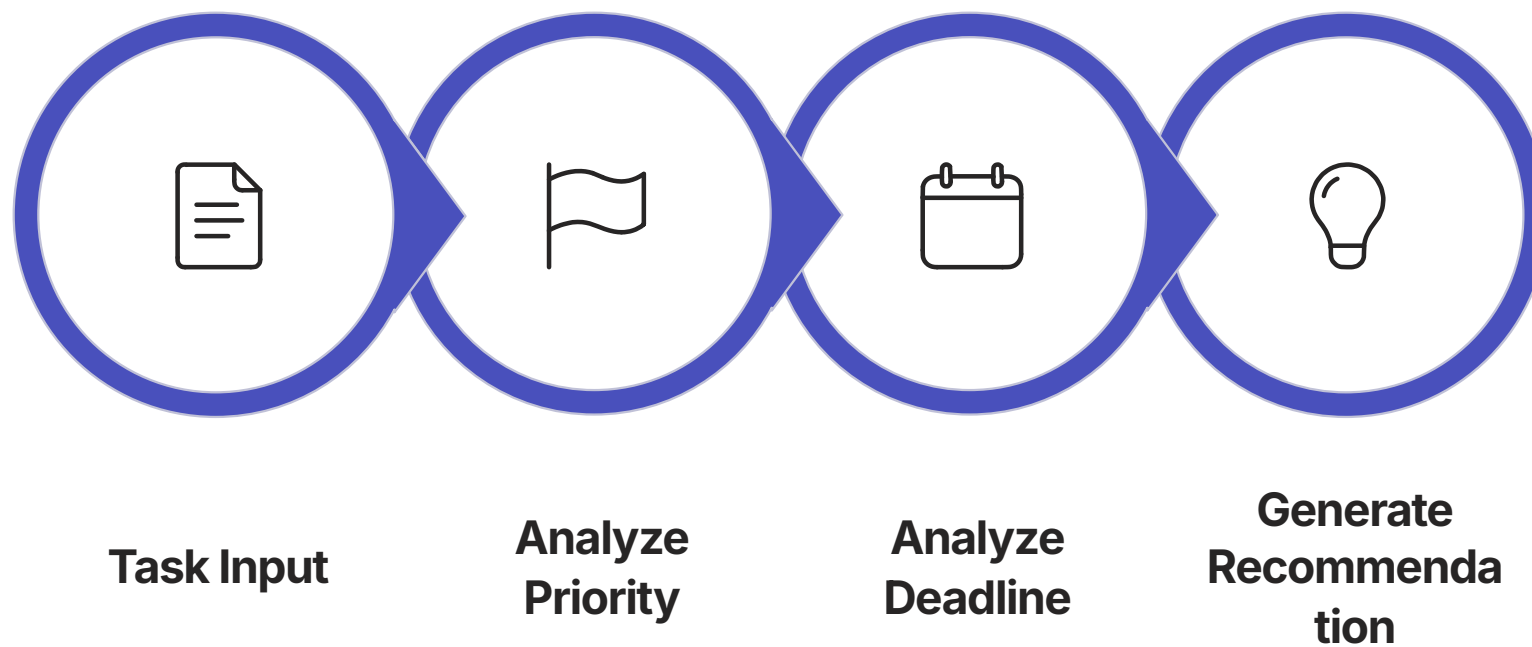
Study Planner Table

- **id**: Unique identifier for the study plan entry.
- **user_id**: Foreign key linking to the Users table.
- **task_id**: Foreign key linking to the Tasks table.
- **study_date**: Date for the planned study session.
- **time_slot**: Allocated time slot for study (e.g., 'morning', 'afternoon', 'evening').

Key Relationships

- **One User → Many Courses**: A single user can enroll in and manage multiple courses.
- **One Course → Many Tasks**: Each course can have numerous associated tasks.
- **One User → Many Study Plans**: A user can have multiple entries in their personalized study planner, with each entry referring to a specific task.

AI Recommendation Engine



This diagram illustrates the core process of the AI Recommendation Engine, transforming raw task inputs into actionable study recommendations.

Priority Scoring:

- High = 3
- Medium = 2
- Low = 1

Deadline Urgency:

- Today = Highest Priority
- 1–3 Days = High
- 4–7 Days = Medium
- More than 7 Days = Low

Output:

- Recommended study tasks
- Study duration
- Completion order

Testing & Results

Testing Summary

- Unit Testing (Authentication APIs): Passed
- Integration Testing (Frontend + Backend): Passed
- Database Testing (CRUD Operations): Passed
- API Testing (Postman): Passed

Key Results

- User authentication system fully functional
- Course management working seamlessly
- Task management with deadline tracking
- Daily planner with AI recommendations
- Dashboard analytics operational
- All API endpoints tested and validated

Benefits & Impact

For Students

- Better time management
- Increased productivity
- Improved learning progress
- Reduced missed deadlines
- Personalized study guidance

For Learning

- Organized course structure
- Clear task prioritization
- Data-driven insights
- Progress tracking
- Intelligent recommendations

For Future

- Scalable architecture
- Ready for mobile app
- AI chatbot integration
- Email notifications
- Performance prediction

Future Enhancements

We have several exciting features planned to further enhance the platform and provide an even richer learning experience for students.

- AI Chatbot for student support
- Mobile Application (iOS & Android)
- Email Notifications for deadlines
- Google Calendar Integration
- Machine Learning-based Performance Prediction
- Voice Assistant Support
- Gamification and Leaderboards
- Advanced Analytics Dashboard
- Peer Collaboration Features
- Integration with Learning Management Systems

These enhancements will focus on expanding the platform's capabilities, improving user engagement, and integrating with external tools to create a comprehensive and intuitive learning ecosystem.